



**ALLIANCE
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GRADE A+**
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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No: 6

Title: Creating HR Analytics Dashboards

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Experiment No: 6

Creating HR Analytics Dashboards

Github Link: <https://github.com/Priyanshnolkha/Data-Analytics-For-Business-Projects/tree/main/Experiment%206%20HR%20Analytics%20Dashboard>

Problem Statement:

You are an HR Data Analyst in a multinational organization. The HR department wants to monitor employee demographics, workforce distribution, hiring trends, employee attrition, job satisfaction, and performance through a single dashboard. Design an interactive HR analytics dashboard that supports data-driven workforce planning, employee retention strategies, and human resource management decisions.

Methodology :

Data Visualization (Creating Dashboard With Changing Values):

In my dashboard, I arranged all KPI cards, charts, gauges, and slicers into a single interactive dashboard.

The dashboard includes:

- Total Employees KPI

- Total Attrition KPI

- Attrition Rate KPI

- Average Monthly Income KPI

- Attrition by Department

- Attrition by Job Role

- Attrition by Age Group

- Attrition by Gender

- Education Field Distribution

- Business Travel vs Attrition

- Performance Rating Gauge

- Interactive Slicers

In the dashboard, it enables users to analyze employee demographics, workforce distribution, and attrition trends efficiently and supports data-driven HR decision-making.

Dashboard Images :

Image 1st:

- It displays the overall HR analytics dashboard without any filters.
- It Shows Total Employees (1.47K), Total Attrition (237), and Attrition Rate (16.12%) along with workforce and attrition analysis..
- It provides a complete overview of business performance using interactive charts and slicers.

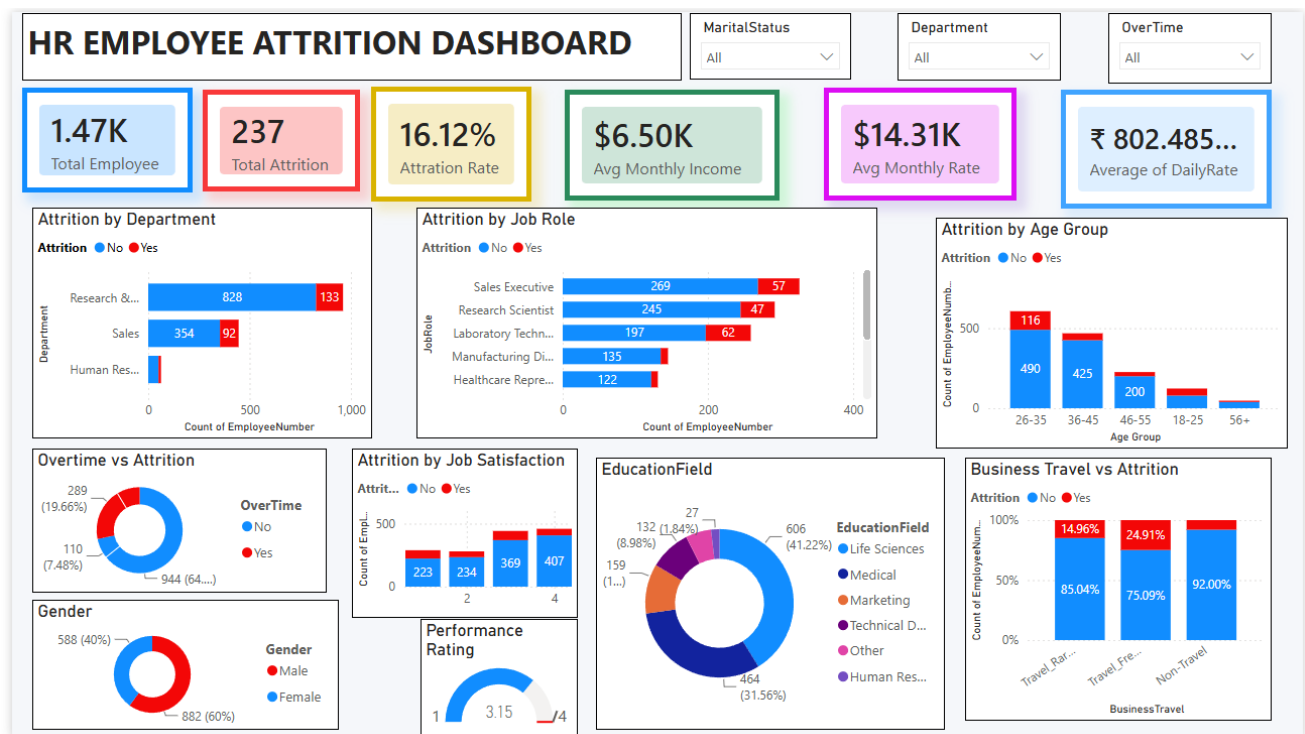


Image 2nd:

In dashboard after selecting the Department Column Chart

- After selecting the Research & Development bar in the *Attrition by Department* chart, all visuals are filtered to display data for the selected department only.
- The KPI cards are updated to Total Employees = 828, Total Attrition = 0, Attrition Rate = 0.00%, and Avg Monthly Income = \$6.63K, while the other charts show information related to the selected department.
- This interaction helps analyze workforce distribution and employee attrition within a specific department.

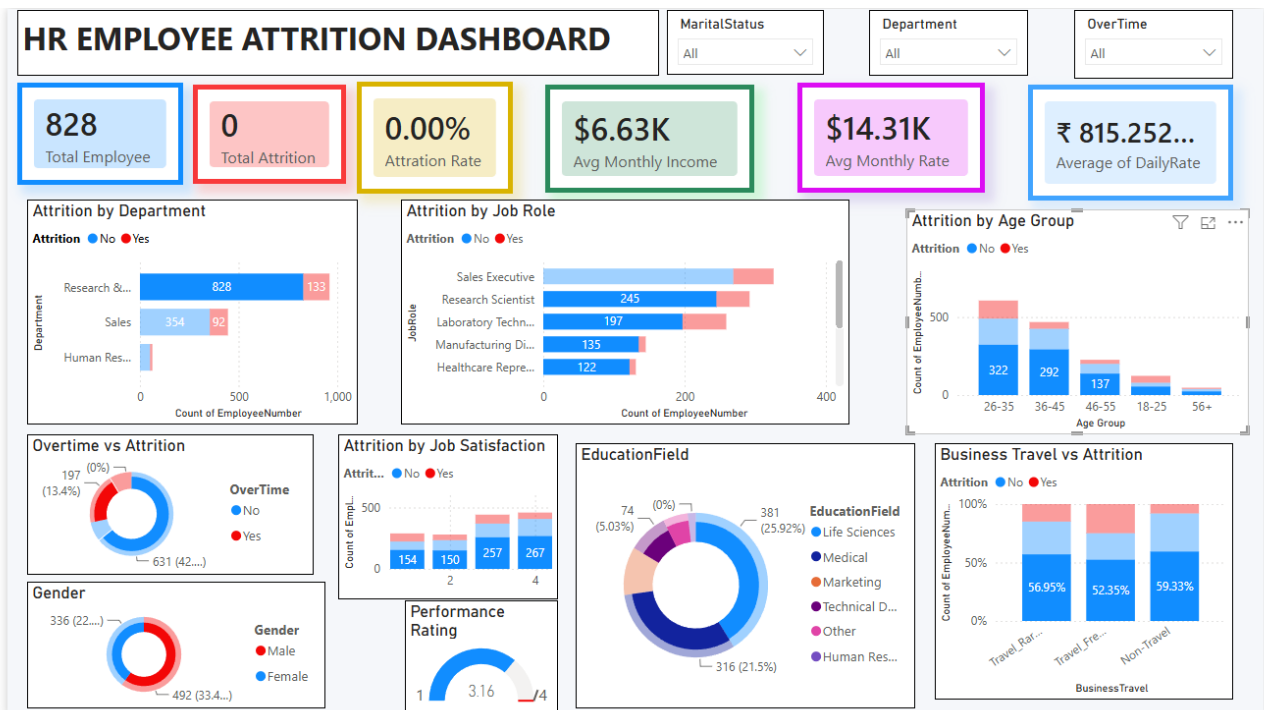
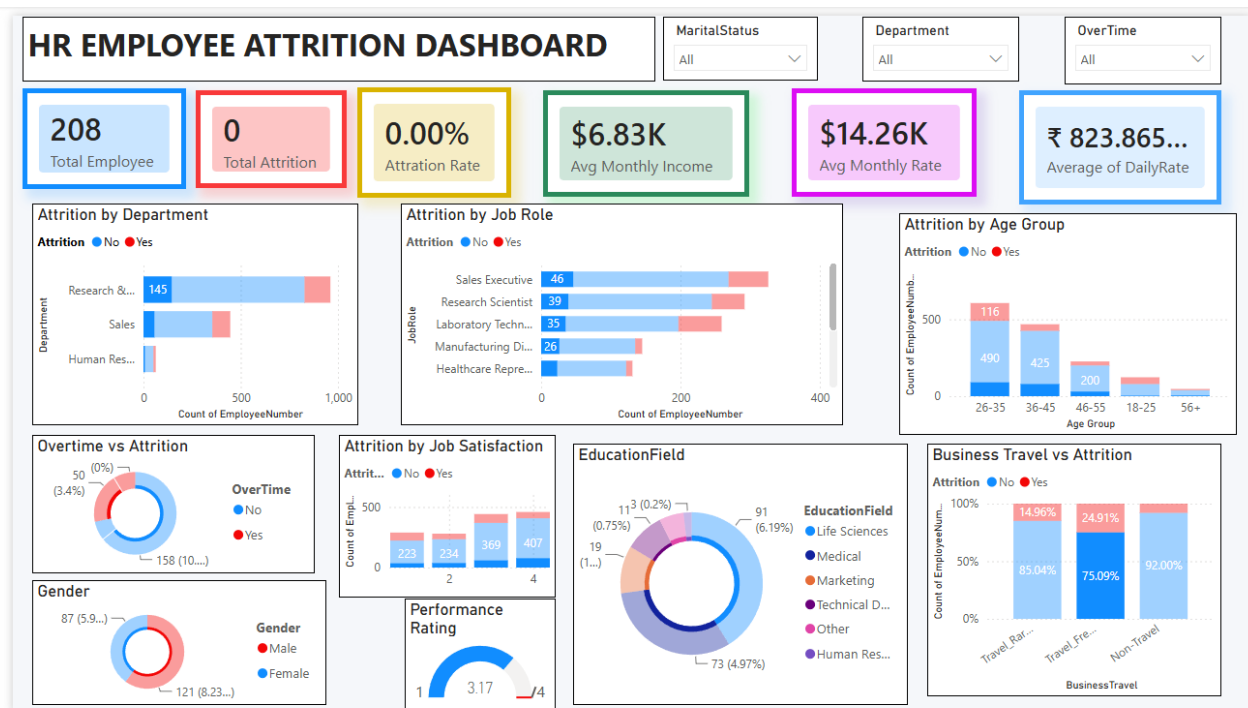


Image 3rd:

In dashboard after selecting the Job Role Bar Chart

- After selecting the Sales Executive bar in the *Attrition by Job Role* chart, all visuals are filtered to display data for the selected job role only.
- The KPI cards are updated to Total Employees = 208, Total Attrition = 0, Attrition Rate = 0.00%, and Avg Monthly Income = \$6.83K, while the other charts show information related to the selected job role.
- This interaction helps analyze workforce distribution, employee demographics, and attrition patterns for a specific job role.



DAX Images :

Image 1st:

- Calculates the average monthly income of all employees using the AVERAGE() function.
- Used in the dashboard to display the Avg Monthly Income KPI (\$6.50K).

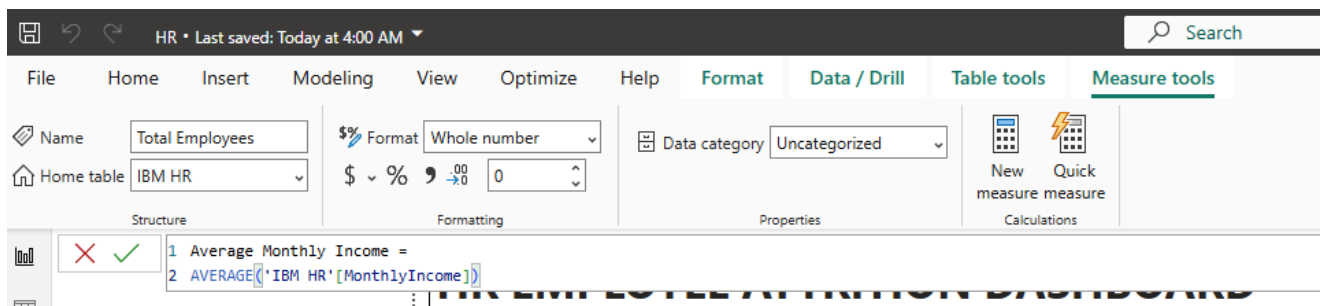


Image 2nd:

- Calculates the total number of employees in the organization using the employee records.
- Used in the dashboard to display the Total Employees KPI (1.47K).

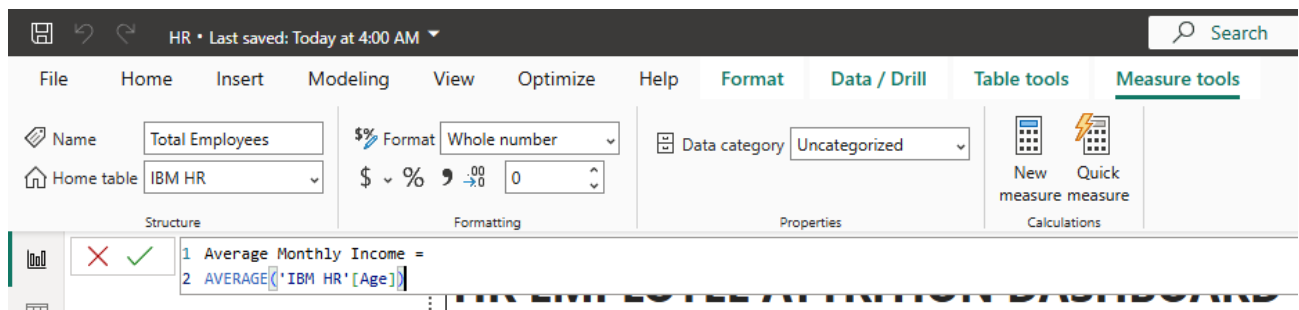
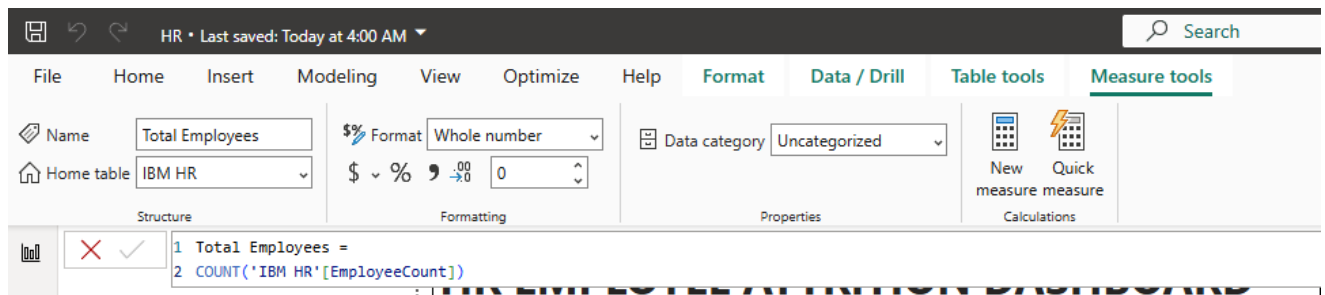


Image 3rd:

- Calculates the total number of employees in the organization using the COUNT() function on the EmployeeCount column.
- Used in the dashboard to display the Total Employees KPI (1.47K) and support workforce analysis.



Business Insights:

- The organization has 1.47K employees with 237 attritions, resulting in an Attrition Rate of 16.12%.
- The Research & Development Department has the highest employee count and attrition.
- Employees aged 26–35 years show the highest attrition.
- Sales Executives, Research Scientists, and Laboratory Technicians experience higher employee turnover.
- Interactive slicers (Marital Status, Department, and Overtime) help analyze workforce trends and support data-driven HR decisions.

Conclusion:

The IBM HR Analytics Employee Attrition dataset was successfully imported into Microsoft Power BI and transformed into an interactive HR analytics dashboard using KPI cards, bar charts, column charts, donut charts, gauges, and slicers. The dashboard effectively analyzed employee demographics, workforce distribution, attrition trends, job roles, and departmental performance, providing valuable HR insights. Overall, the interactive dashboard supports data-driven workforce planning and employee management by enabling HR professionals to monitor and evaluate employee attrition efficiently.

